



SPARK PERFORMANCE TUNING

A Practical Checklist for Diagnosing Slow Jobs

Spark UI

Partitions

Joins

AQE

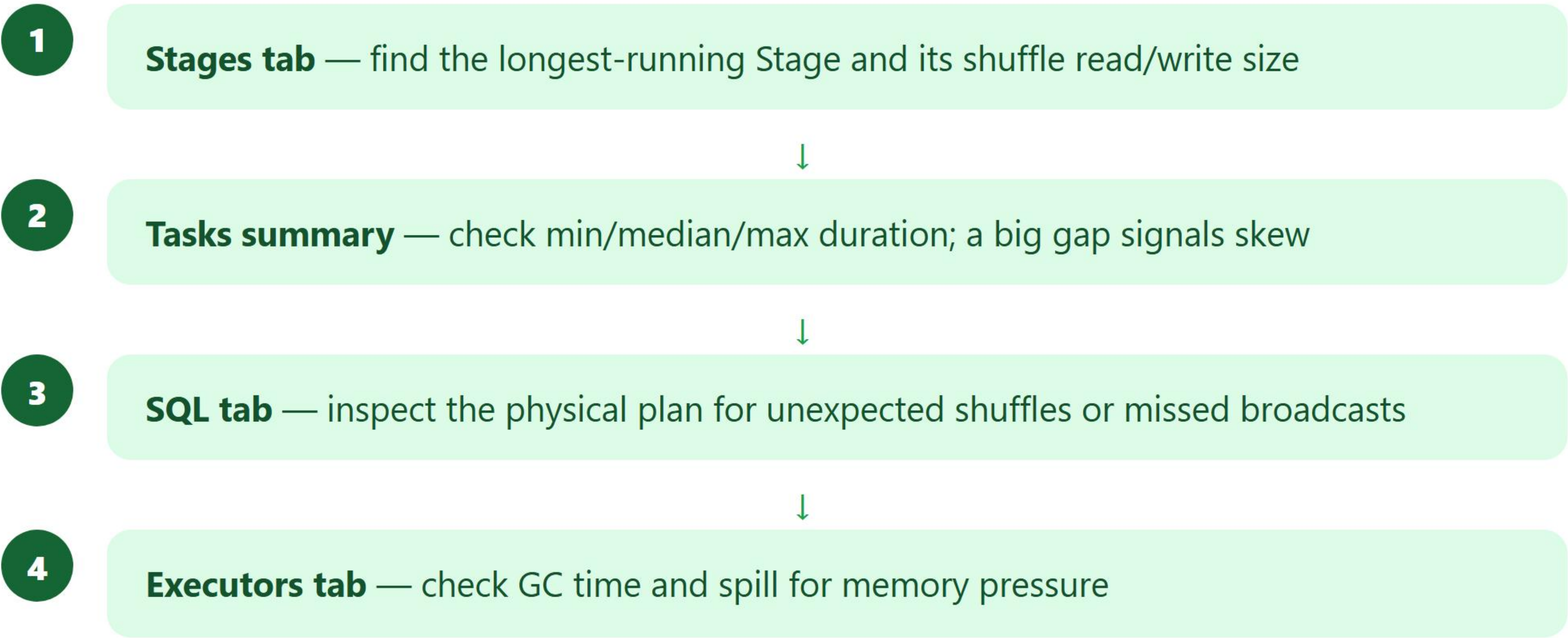
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U **START WITH THE SPARK UI**

Every tuning decision should start from evidence in the Spark UI — guessing wastes time and can make things worse.

Where to Look First



Interview Tip: If you can't point to a specific Stage or metric, you're not ready to change a config yet.

L THE HIGH-LEVERAGE CHANGES

WHERE MOST WINS COME FROM

Partitioning

Caching

Broadcast Joins

File Format

AQE

What to Actually Change

1

Right-size **spark.sql.shuffle.partitions** — too few starves parallelism, too many adds overhead



2

Cache only DataFrames reused across multiple Actions, then **.unpersist()** when done



3

Force **broadcast joins** for small dimension tables Catalyst underestimates



4

Use columnar formats (**Parquet/ORC**) with predicate pushdown over CSV/JSON

Interview Tip: These five levers fix the majority of real-world slow jobs — most other tuning is diminishing returns.

A LET SPARK RE-PLAN AT RUNTIME

AQE re-optimizes the physical plan **mid-Job** using actual shuffle statistics instead of only upfront estimates.

What AQE Fixes Automatically

1

Coalesces small post-shuffle partitions — no manual `shuffle.partitions` tuning



2

Switches Sort-Merge → **Broadcast Join** if runtime stats show one side is small



3

Splits skewed partitions into smaller sub-tasks automatically



4

Enabled by default since Spark 3.2 — `spark.sql.adaptive.enabled`

Interview Tip: AQE turns compile-time guesses into runtime facts — mention it whenever discussing join or partition tuning.

X FREQUENT MISTAKES TO AVOID

Most performance regressions trace back to one of these four habits.

What Quietly Kills Performance



.collect() on large DataFrames — pulls everything to the Driver, risking OOM



Python UDFs instead of built-in functions — breaks Whole-Stage CodeGen



Excessive **.cache()** calls that evict useful blocks and fragment Storage Memory



Reading **many small files** — creates thousands of tiny partitions and Task overhead

Interview Tip: Prefer built-in Spark SQL functions over UDFs whenever possible — UDFs are a black box to Catalyst.

Save this slide before your next Data Engineering interview

LEVER	FIXES	KEY INTERVIEW FACT
Spark UI	Diagnosis	Always confirm the bottleneck Stage first
Partitioning	Parallelism / skew	Right-size shuffle.partitions
Broadcast Join	Shuffle elimination	Force via broadcast(df) hint
AQE	Runtime re-planning	Enabled by default since Spark 3.2
Avoid UDFs/collect()	CodeGen & OOM	Prefer built-ins, avoid pulling data to Driver

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